CCNA REVIEW NOTES

****DISCLAIMER: This is not an official CCNA study guide. All topics listed are in reference to the CCNA Exam Topics list posted to Cisco.com. This document assumes the reader has already studied the foundational concepts and should be used for review only.**

IP Addressing

IPv4 / IPv6 Multicast Addresses:

Address For	IPv6 Address	IPv4 Address	Note
All Nodes	FF02::1	224.0.0.1	NDP RA messages
All Routers	FF02::2	224.0.0.2	HSRP V1, NDP RS messages
All OSPF Routers	FF02::5	224.0.0.5	
ALL OSPF DR/BDRs	FF02::6	224.0.0.6	
ALL RIP Routers	FF02::9	224.0.0.9	
ALL EIGRP Routers	FF02::A	224.0.0.10	
VRRP		224.0.0.18	
GLBP		224.0.0.102	HSRP V2

IPv4 Public Classes

Class A	0.0.0.0	127.255.255.255
Class B	128.0.0.0	191.255.255.255
Class C	192.0.0.0	223.255.255.255
Loopback	127.0.0.0	127.255.255.255

IPv4 Private Address Ranges

10.0.0.0	10.255.255.255 /8
172.16.0.0	172.31.255.255 /12
192.168.0.0	192.168.255.255 /16

IPv6 Address Types

Name	Range	Note
Global Unicast	Any that’s not reserved	Public, globally unique
Unique Local	FC00::/7 (Starts with FD)	Like an IPv4 private address
Link-Local	FE80::/10	Auto generated with EUI-64
Multicast	FF00::/8 (Usually FF02)	One-to-many communication
Anycast	Any that's not reserved	When config., specify anycast

IPv6 Multicast Scopes

Interface-local	FF01
Link-Local	FF02
Site-local	FF05
Organization-local	FF08
Global	FF0E

EUI-64 Generation Steps

1.)	Divide MAC address in half
2.)	Insert FFFE in the middle
3.)	Invert 7th bit (2nd hexadecimal in the MAC address) Ex: 8 -> A

Solicited-node Multicast Address

ff02::1:ff [last 6 hex digits of unicast address]
Example: 2001:db8::1:f2a:4fff:fea3:91b1 → ff02::1:ffa3:91b1

NDP Message Types

RS (Router Solicitation)	ICMPv6 Type 133	FF02::2, SLAAC messages
RA (Router Advertisement)	ICMPv6 Type 134	FF02::1, SLAAC messages
NS (Neighbor Solicitation)	ICMPv6 Type 135	DAD messages
NA (Neighbor Advertisement)	ICMPv6 Type 136	DAD messages

MAC Addresses

Broadcast	ffff.ffff.ffff
HSRP V1 Virtual Format	0000.0c07.acxx (xx = group number)
HSRP V2 Virtual Format	0000.0c9f.fxxx (xxx = group number)
VRRP Virtual Format	0000.5e00.01xx (xx = group number)
GLBP Virtual Format	0007.b400.xxyy (xx = group, yy = router)
Standard STP BPDU's Destination MAC	0180.c200.0000
PVST+ BPDU's Destination MAC	0100.0ccc.cccd
LLDP Destination MAC	0180.c200.000e
CDP Destination MAC	0100.0ccc.cccc

Cable Topics

Copper Ethernet Cable Standards

10 Mbps	Ethernet	802.3i	10 Base-T	100m
100 Mbps	Fast Ethernet	802.3u	100 Base-T	100m
1 Gbps	Gigabit Ethernet	802.3ab	1000 Base-T	100m
10 Gbps	10 Gig Ethernet	802.3an	10G Base-T	100m

Pin Pairs

Device	Transmits on	Received on
Firewall	1 and 2	3 and 6
Router	1 and 2	3 and 6
PC	1 and 2	3 and 6
Switch	3 and 6	1 and 2

Fiber Cable Standards

1000 Base-LX	802.3z	1 Gbps	Multi or Single-Mode	550m (MM) 5 KM (SM)
10G Base-SR	802.3ae	10 Gbps	Multimode	400 m
10G Base-LR	802.3ae	10 Gbps	Single-mode	10 km
10G Base-ER	802.3ae	10 Gbps	Single-mode	30 km

Serial Standards (WAN)

T1 (slowest)	1.544 Mbps	North America / Japan
T2	6.312 Mbps	North America / Japan
T3 (fastest)	44.736 Mbps	North America / Japan
E1	2.048 Mbps	Europe
E2	8.448 Mbps	Europe
E3	34.368 Mbps	Europe

Cable Terminators

Ethernet	RJ45
Fiber	ST
Point-to-Point Serial	DB-9
Coaxial	BNC
POTS	RJ-11

Duplex Errors

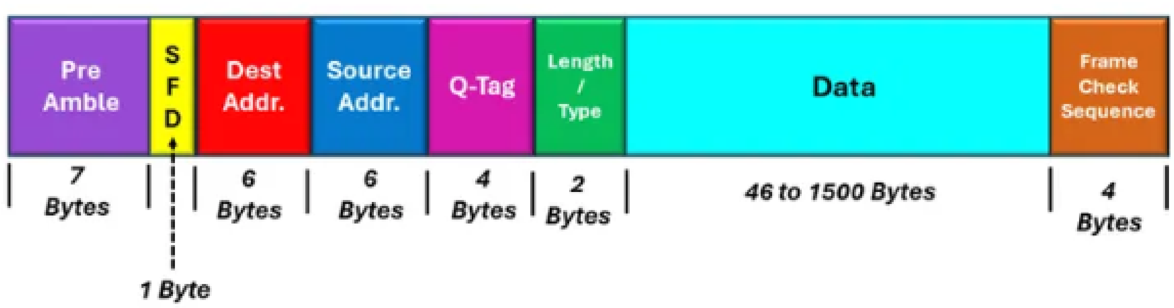
Symptoms	Runts, FCS and CRC errors, Alignment errors, Late collisions

If Duplex Autonegotiation is Off on One Device

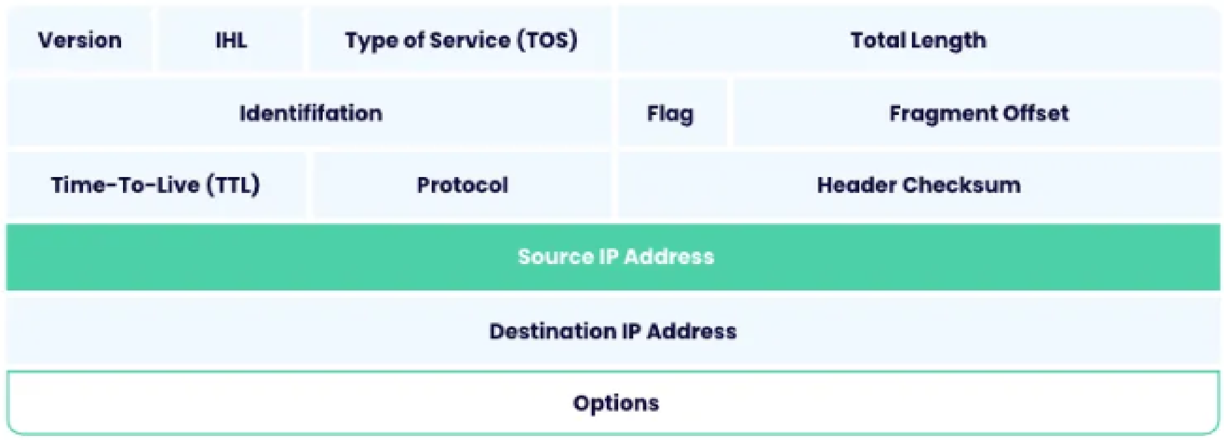
Speed	Other device will try to sense speed, if it can't; it will operate at slowest supported speed
Duplex	10 or 100 Mbps = Half duplex 1000Mbps (1Gbps) or More = Full duplex

Headers / Segments

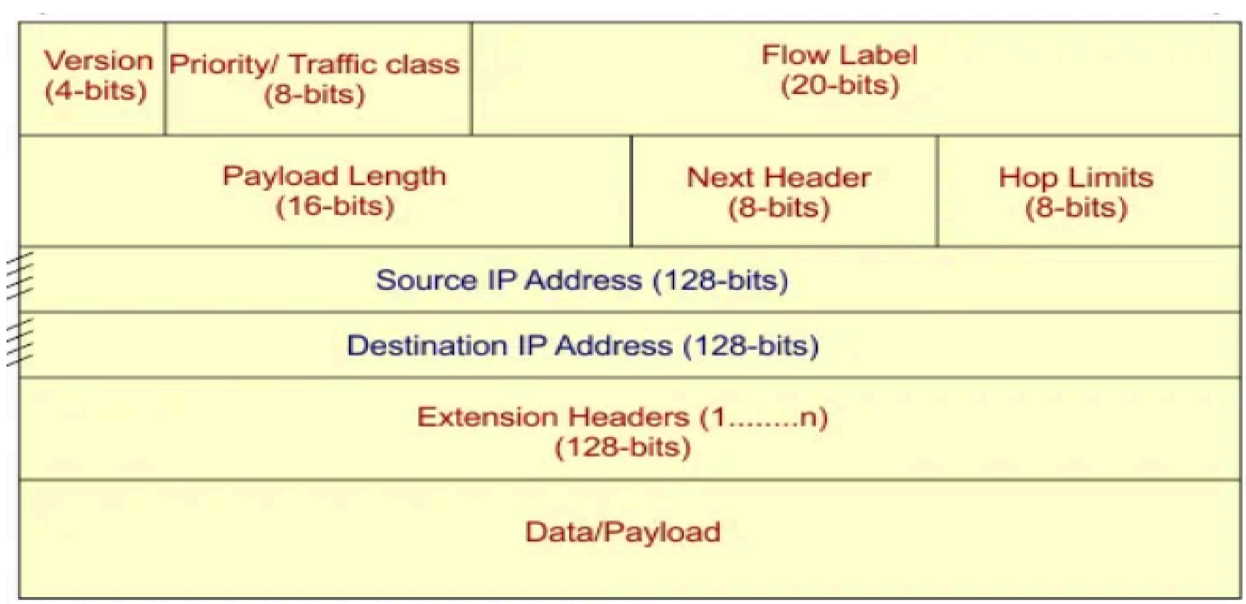
Ethernet Header



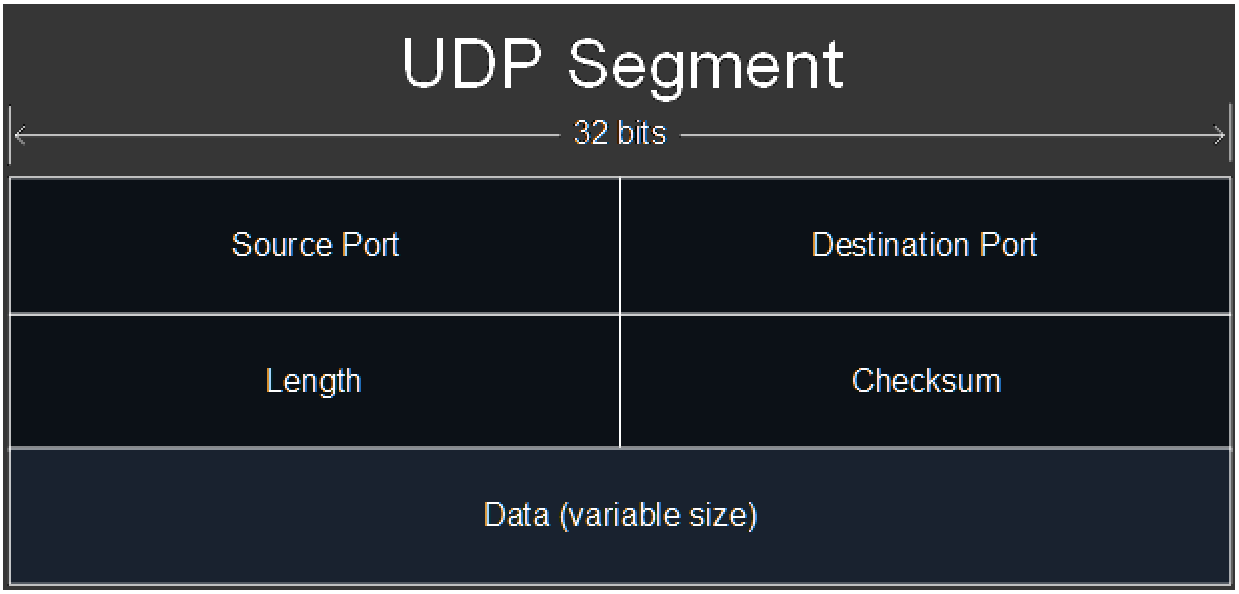
IPv4 Header



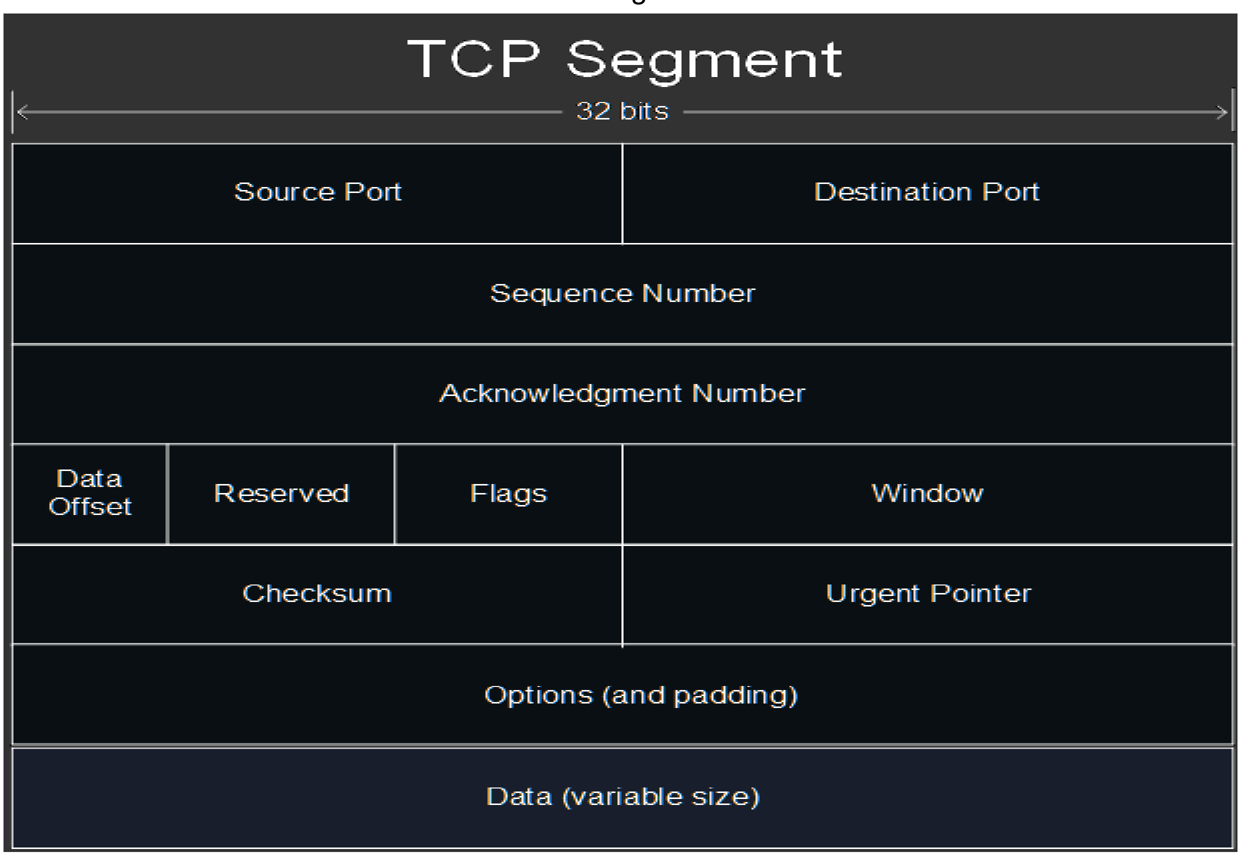
IPv6 Header



UDP Segment



TCP Segment



Port numbers

TCP	UDP	TCP & UDP
FTP Data (20)	DHCP Server (67)	DNS (53) (Usually UDP)
FTP Control (21)	DHCP Client (68)	DNS uses TCP > 512 bytes
SSH (22)	TFTP (69)	
Telnet (23)	NTP (123)	
SMTP (25)	SNMP Agent (161)	
TACACS+ (49)	SNMP Manager (162)	
HTTP (80)	Syslog (514)	
POP3 (110)	RADIUS (1812)	
HTTPS (443)	CAPWAP Control (5246)	
	CAPWAP Data (5247)	

Layer 4 Port Ranges

Well-Known Ports	0 - 1023
Registered Ports	1024 - 49151
Ephemeral Ports	49152 - 65535

IPv4 Header Protocol Numbers

ICMP	1
TCP	6
UDP	17
EIGRP	88
OSPF	89

Administrative Distances

Route Protocol / Type	AD
Directly Connected	0
Static	1
External BGP (eBGP)	20
EIGRP	90
IGRP	100
OSPF	110
IS-IS	115
RIP	120
EIGRP (external)	170
Internal BGP (iBGP)	200
Unusable Route	255

Random Protocol Topics

STP/RSTP Root Bridge Election

1.) Lowest Bridge Priority
2.) Tiebreaker: Lowest MAC Address

STP/RSTP Root Port Election

1.) Lowest Root Cost (bandwidth)
2.) Lowest Bridge ID (Bridge Priority + MAC Address)
3.) Lowest Neighbor Port ID

STP Standards

Classic STP	802.1D
Multiple STP	802.1s
PVST+	Cisco upgrade to 802.1D
RSTP	802.1w
Rapid PVST+	Cisco upgrade to 802.1w

RSTP Port Roles

Root	To Root Bridge (forwarding)
Designated	From Root Bridge and on Some Other Bridges (forwarding)
Alternate	Backup to Root Port (discarding) superior BPDU from other switch
Backup	Usually for Hubs (discarding) superior BPDU from same switch

Etherchannel

PAgP (Cisco Proprietary)	LACP (802.3ad Standard)
Channel-protocol pagp	Channel-protocol lacp
Desirable, Auto	Active, Passive

Etherchannel Commands

Where / Type of Command	Term used	Examples
Global configuration (config)#	port-channel	Port-channel load-balance Interface port-channel <*num*>
Interface configuration (config-if)#	channel-group	Channel-group <*num*> mode Channel-protocol <*protocol*>
Show commands #	etherchannel	Show etherchannel summary

Layer 3 Etherchannel Configuration

1.) Enable routing (L3 switches)	(config)# ip routing
2.) Make SVI, assign to channel-group	(config-if)# no switchport (config-if)# channel-group 1 mode <*mode*>
3.) Enter config. For that port-channel	(config)# int port-channel 1
4.) Assign IP and enable	(config-if)# ip address 192.168.1.1 255.255.255.0 (config-if)# no shut

CDP / LLDP Info

Name	Timer	Holdtime	Note
CDP (Cisco)	60 sec	180 sec	Shows VTP in detail command, enabled by default
LLDP (802.1ab)	30 sec	120 sec	Tx and Rx on interfaces separately enabled, default re-initialization timer is 2 seconds

CDP Show Commands

Show cdp neighbors	Device ID, Local Int, Holdtime, Capabilities, Product Number, Neighbor Port ID
Show cdp neighbors detail	Everything in normal command + IP Address, Software Version, Native VLAN, VTP Domain
Show cdp entry <*name*>	Like Detail, but for a specific device in the network

OSPF DR/BDR Election

1.) Highest Priority (1-255), Second Highest becomes BDR (default is 1)
2.) Tiebreaker: Highest Router ID

OSPF Router ID Selection

1.) Manually Configured Router ID
2.) Highest Loopback Interface IP Address
3.) Highest Physical Interface IP Address

OSPF Neighbor States

Down	No Neighbors yet
Init	Hello received, own Router ID not in packet
2-Way	Received Hello with own Router ID, DR and BDR elected
Exstart	Choose Master/Slave, Higher Router ID becomes master
Exchange	Exchange DBDs
Loading	Send LSR and LSU to update their LSDBs
Full	Adjacency, DROthers only form Full with DR and BDR

OSPF Network Types

Broadcast	Point-to-Point	Nonbroadcast
DR and BDR elected	No DR or BDR	DR and BDR elected
Multicast	Multicast	Unicast
Ethernet and FDDI Interfaces	HDLC and PPP Interfaces	Frame Relay and X.25 Interfaces
Hello 10, Dead 40	Hello 10, Dead 40	Hello 30, Dead 120
Dynamic Adjacencies	Dynamic Adjacencies	Manual Adjacencies (neighbor command)

OSPF LSA Types

Type	Note
Type 1 (Router LSA)	Generated by every router, lists Router ID, networks attached
Type 2 (Network LSA)	Generated by the DR, lists routers attached to the multi-access network
Type 5 (AS-External LSA)	Generated by ASBR, advertises routes to destinations outside the AS

Requirements for OSPF Adjacency

Area Number must match
Interfaces must be in same subnet
OSPF process must not be shutdown
OSPF Router IDs must be unique
Hello and Dead timers must match
Authentication Settings must match

Other Requirements for OSPF Adjacency

IP MTU Settings must match	Become neighbors, but won’t go to Full state
OSPF Network Type must match	Prevents proper adjacency, won’t learn OSPF routes

Random OSPF Configuration Commands

Configure Cost of an Interface	(config-if)# ip ospf cost <*cost*>
Configure the Reference Bandwidth	(config-router)# auto-cost reference-bandwidth <*mbps*>
Enable OSPF Directly on an Interface	(config-if)# ip ospf <*process-id*> area <*area num*>
Enable OSPF Authentication	(config-if)# ip ospf authentication
Configure an OSPF Password	(config-if)# ip ospf authentication-key <*password*>
Set the OSPF Network Type on an Int.	(config-if)# ip ospf network <*type*>

Dynamic Routing Protocol Comparison

OSPF	RIP	EIGRP
Link-state	Distance Vector	Distance Vector
Router ospf [process-ID]	Router rip	Router eigrp [AS-number]
Network *<ip>* *<wildcard>* area *<num>*	Network *<ip>*	Network *<ip>* *<wildcard>*
Does not have auto-summary	Auto-summary enabled, default (should turn off)	Auto-summary enabled, default (should turn off)
Area must match, process-ID does not have to match between routers	n/a	AS-number must match between routers
Bandwidth (cost) metric	Hop count metric	Bandwidth (slowest link) + delay (all links) metric
Only ECMP load-balance	Only ECMP load-balance	Supports different cost load-balance (using feasible successor routes)

EIGRP Terms

Successor	Route with lowest metric to destination (best path)
Feasible Successor	Alternate route which meets the feasibility condition
Feasible Distance	The router’s own metric to the destination
Reported Distance	The neighbor’s metric value to the destination

Feasibility Condition

A route is considered a Feasible Successor if its Reported Distance is lower than the Successor route’s Feasible Distance.

FHRP Protocols

HSRP	Active, Standby (Cisco) (Load-Balance per VLAN) (ONE Virtual MAC)
VRRP	Master, Backup (Industry-standard) (Load-Balance per VLAN)
GLBP	AVG, AVF (Cisco) (Same VLAN Load-Balancing via AVFs)

HSRP Active Router Election

1.) Highest Priority (Default 100)
2.) Highest IP Address

SNMP Messages

Message Type	SNMP Message Name	Note
Read	Get, GetNext, GetBulk	Sent by NMS
Write	Set	Sent by NMS
Notification	Trap (unreliable), Inform	Sent by Managed Device
Response	Response	Response to a message

Syslog Severity Levels

0	Emergency	Ex: Frozen or force restart the OS
1	Alert	Ex: Failure of database auth system- halts regular operations
2	Critical	Ex: Hardware component failure
3	Error	Ex: Auth failures, DNS lookup failures
4	Warning	Ex: High CPU load, certificate expiration
5	Notice/Notification	Ex: Interface Up/Down messages, critical config. changes
6	Informational	Ex: Successful logon or log off events, process completions
7	Debugging	Ex: System diagnostics or troubleshooting

Remember: **E**very **A**wesome **C**isco **E**ngineer **W**ill **N**eed **I**cecream **D**aily

NTP Configuration

Ntp server *<ip>*	Sets server that device gets its time from, static client/server mode
Ntp master	Makes the device an NTP server (default stratum of 8), server mode
Ntp peer *<ip>*	Symmetric active mode - peers with other devices
Ntp broadcast client	Listens for broadcast messages on that port from NTP servers

DNS Configuration Commands

Configure the DNS server the router will send queries to	Ip name-server <*ip*>
Configure the domain of the router	Ip domain name <*name*>
Configure the router to act as a DNS server	Ip dns server
Enable the router to perform DNS queries	Ip domain lookup

Static NAT Configuration

Static NAT	Ip nat inside source static <*inside local*> <*inside global*>
Apply to Interfaces	Ip nat inside, ip nat outside

Dynamic NAT/PAT Configuration

Create Pool	Ip nat pool <*name*> <*start ip*> <*end ip*> netmask <*mask*>
Create ACL	Access-list <*acl number*> permit <*source network*> <*wildcard*>
Config. NAT/PAT	Ip nat inside source list <*acl num*> pool <*pool name*> [overload]
Apply to Interfaces	Ip nat inside, ip nat outside

DHCP Messages

Message	Sent by	Note
DISCOVER	Client	Broadcast
OFFER	Server	Broadcast or Unicast
REQUEST	Client	Broadcast
ACK	Server	Broadcast or Unicast
RELEASE	Client	Unicast, release a DHCP IP address
NAK	Server	Server declines a client request
DECLINE	Client	Declines a server offered DHCP IP address

DHCP Snooping Message Checks

DHCP Messages Received on Untrusted Port	Snooping Checks
DISCOVER, REQUEST	Source MAC to CHADDR for match
RELEASE, DECLINE	Source IP to Interface for match (binding)

DHCP Snooping Configuration List

Enable DHCP Snooping Globally, and on VLANs	(config)# ip dhcp snooping (config)# ip dhcp snooping vlan <*vlan-ids*>
*For L2 Switches (Access switches): disable Option 82	(config)# no ip dhcp snooping information option
Trust interfaces connected to servers / upstream networking devices	(config-if)# ip dhcp snooping trust
Enable Rate Limiting (Best practice)	(config-if)# ip dhcp snooping limit rate <*pps*>

DAI Message Checks

Sender MAC to Sender IP in DHCP Snooping Binding Table and/or ARP ACL

DHCP Snooping vs DAI Comparison

DHCP Snooping	DAI
Needs 2 Commands to enable (1 global, 1 on a VLAN) (ip dhcp snooping) (ip dhcp snooping vlan [vlan-id])	Only needs one command to enable (ip arp inspection vlan [vlan-id])
Rate limiting disabled by default	Rate limiting enabled by default (15 pps)

Port-Security Defaults

After the switchport port-security command on an interface:
Max of 1 MAC address allowed (Dynamically learned)
Violation mode: Shutdown
Static MAC Address Aging Disabled
No Errdisable Recovery

*Remember: Port-Security only allowed on Static Access or Trunk ports, DTP must be off

Port-Security Violation Modes

Shutdown	Restrict	Protect
Errdisable Interface if violated	Interface stays up, drops unknown packets	Interface stays up, drops unknown packets
Generates SNMP/Syslog messages	Generates SNMP/Syslog messages	Does not generate SNMP/Syslog messages
Increments violation counter	Increments violation counter	Does not increment violation counter

SSH Config List

1.) Hostname (hostname <*name*>)
2.) Domain Name (ip domain-name <*name*>)
3.) Generate Key Pair (crypto key generate rsa)
4.) Username and Password (username <*name*> secret <*password*>)
5.) VTY config (line vty 0 15) (transport input ssh) (login local)

ACL Types

ACL Type	Numbered Range	Should Be Configured Closest To
Standard	1-99, 1300-1999	Closest to Destination
Extended	100-199, 2000-2699	Closest to Source

Apply an ACL

Apply ACL to VTY Lines	access-class <*name/number*> [in/out] ← IPv4 ipv6 access-class <*name/number*> [in/out] ← IPv6
Apply ACL to Interface	ip access-group <*name/number*> [in/out] ← IPv4 ipv6 traffic-filter <*name/number*> [in/out] ← IPv6
Apply ARP ACL to DAI	ip arp inspection filter <*acl name*> vlan <*vlan num*>

Security

Password Configuration Commands

Enable secret <*password*>	Configures and encrypts a clear-text password for privileged exec mode (md5)
Enable password <*password*>	Configures a clear-text password for privileged exec mode
Service password-encryption	Enables global password encryption
Password 7 <*hash*>	Configures an encrypted vty login password (type 7)
Enable secret 5 <*hash*>	Configures a previously encrypted password for enable mode (md5) (Must plug in a <u>hash</u>, not the actual password)
Enable secret level <*num*> <*password*>	Configures a password to access a specified privilege level (0-15)
Username <*name*> [**privilege** (0-15)] {**password** (encryption-type) <*password*>}	Configure a user account with a certain privilege level. *Can replace <u>password</u> with <u>secret</u> for md5 encryption

Password Encryption Levels

Type 0 (clear-text)	Enable password
Type 5 (md5)	Enable secret
Type 7 (weak)	Service password-encryption
Type 9 (scrypt)	User <*name*> algorithm-type scrypt secret <*password*>

Privilege Levels

15	Privileged Exec mode, most commands issued here
1	User Exec mode, default nonprivileged login prompt level
0	5 commands default - disable, enable, exit, help, logout

AAA Authentication Configuration

<table>
<tr><td>Create Model</td><td>(config)# aaa new-model</td></tr>
<tr><td>(As Needed) Define RADIUS Server</td><td>(config)# radius-server host <ip> key <password></td></tr>
<tr><td>Create AAA list</td><td>(config)# aaa authentication login {default | list-name} <method(s)></td></tr>
<tr><td>Apply to Console/Vty lines</td><td>(config-line)# login authentication {default | list-name}</td></tr>
</table>

TACACS+ and RADIUS

TACACS+ (Cisco)	RADIUS (Industry)
TCP 49	UDP 1645, 1812
Encrypts entire packet	Doesn't encrypt entire packet
Can authorize a user to a subset of CLI Commands	Does not support subset authorization function

Random Timers (Defaults)

OSPF LSA max age	30 minutes
Err-disable recovery interval	5 minutes
STP Timers	Hello = 2 seconds, Max Age = 20 seconds Forward Delay = 15 seconds
RSTP Timers	Hello = 2 seconds, Keepalive = 3 BPDUs
Dynamically learned MAC Addresses	Aging time = 5 minutes
HSRP Timers	Hello = 3 seconds, Holdtime = 10 seconds

QoS Topics

Methods to Classify Traffic

PCP (also known as CoS)	802.1Q VLAN tag (Ethernet Header) Only on Trunk links, or Access ports with Voice VLAN (PC traffic not tagged)
DSCP	Layer 3 QoS (IP Header)

DSCP Markings

DF (Default Forwarding)	Best Effort (value of 0)
EF (Expedited Forwarding)	Usually Voice (value of 46)
AF (Assured Forwarding)	AF*xy* (x=priority, y=drop precedence)
CS (Class Selector)	8 values (backwards compatibility with IPP)

Assured Forwarding Values

AF41 (Best Treatment)	AF42	AF43
AF31	AF32	AF33
AF21	AF22	AF23
AF11	AF12	AF13 (Worst Treatment)

RFC Recommendations

Voice Traffic	EF (Expedited Forwarding)
Interactive Video	AF4x
Streaming Video	AF3x
High Priority	AF2x
Best Effort	DF (Default Forwarding)

PoE Standards

Name	Standard	Watts	Powered Wire Pairs
Cisco Inline Power	Cisco	7	2
PoE (type 1)	802.3af	15	2
PoE+ (type 2)	802.3at	30	2
UPoE (type 3)	802.3bt	60	4
UPoE+ (type 4)	802.3bt	100	4

Interactive Audio Recommendations

One-Way Delay	150ms or less
Jitter	30ms or less
Loss	1% or less

VoIP Configuration Commands

Switchport voice vlan dot1p	Separates voice and data traffic, but does not create a separate voice VLAN
Switchport voice vlan none	Disable the voice VLAN, voice and data traffic unseparated
Switchport voice vlan *<num>*	Sends voice traffic tagged in the configured voice VLAN
Switchport voice vlan untagged	Voice traffic goes through the native/default VLAN

WAN Topologies

Metro Ethernet Topologies

Name	Topology Type	Note
E-Line	Point-to-point	Similar to Leased Line
E-LAN	Full Mesh	Acts like a LAN
E-Tree	Hub and Spoke	Remote sites must go through Central site

MPLS Types

MPLS Layer 3 VPN	MPLS Layer 2 VPN
CE routers form OSPF adjacencies with PE routers	CE routers only form OSPF adjacencies with other CE routers
WAN operations, each branch has own IP subnet	WAN is invisible, acts like a giant LAN switch

VPN Types

Site-to-site VPN	Remote-Access VPN
Usually office to office, permanent, multiple devices supported	On demand, one device creates VPN to connect to the enterprise network
IPsec encryption	TLS encryption
IPsec + GRE for tunnels	Forms tunnels through TLS
Does not require additional software	Requires software on end-user devices
Network Device to Network Device	User Device to Network Gateway Device

Wireless Topics

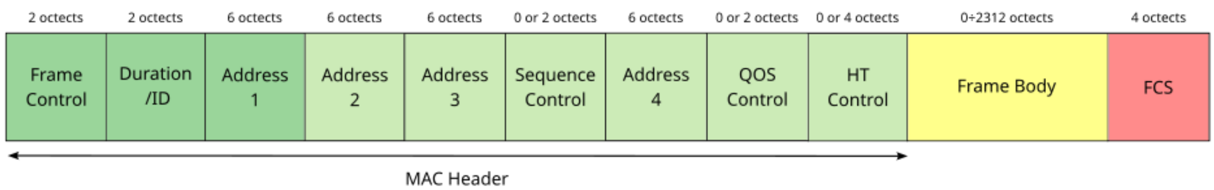
Wireless Standards

802.11	2.4Ghz	2 Mbps	
802.11b	2.4 Ghz	11 Mbps	
802.11a	5 Ghz	54 Mbps	
802.11g	2.4 Ghz	54 Mbps	
802.11n	2.4 / 5 Ghz	600 Mbps	Wi-Fi 4
802.11ac	5 Ghz	6.93 Gbps	Wi-Fi 5
802.11ax	2.4 / 5 / 6 Ghz	4*802.11ac	Wi-Fi 6

Wireless Standards Misc.

802.11v	Network-assisted power savings, uses DMS, client can sleep
802.11r	Fast transition roaming, perform PMK calculation in advance
802.11w	Management Frame Protection, security mechanism
802.11k	Assisted roaming, Client requests a list of candidate transition APs

802.11 Frame Order



802.11 Message Types

Management	Control	Data
Beacon	RTS (Request to Send)	Used to send data packets
Probe request, response	CTS (Clear to Send)	
Authentication	ACK	
Association request, response		

RF Terms

Amplitude	Height of a wave from trough to crest
Wavelength	Length of a wave from crest to crest
Frequency	Number of waves per given time period
Absorption	Wireless signal passes through material, weakening signal
Reflection	Signal bounces off a material, ex. metal
Refraction	Signal goes through a medium, changes direction and speed
Diffraction	Wave encounters object and travels around it, can cause blind spots
Scattering	Material causes signal to scatter in all directions, ex. dust, smog

Service Sets

IBSS	Clients connect directly, no AP needed (ad hoc)
BSS	Clients connect to a single AP
ESS	Multiple APs connect via wired network, each BSS has same SSID, different BSSIDs, different channels
MBSS	APs connect to each other wirelessly, one channel for AP connection, one channel for the BSS

AP Overlap

Acceptable overlap for consistent roaming = 10-15%

Autonomous AP Modes

Repeater	Extend range of BSS
WGB (Workgroup Bridge)	Wireless client of another AP, connect wired client to wireless network
Outdoor Bridge	Long distance, point-to-point

Lightweight AP Modes

Local	Default, sends all client traffic for all WLANs in CAPWAP to the WLC
FlexConnect	Local, but can act Autonomous if connection to WLC goes down
Sniffer	No BSS offer, sends traffic to Wireshark or similar
Monitor	No BSS, to detect and disassociate rogue devices
Rogue Detector	No radio, detects rogue devices on the <u>wired</u> network
SE-Connect	No BSS, dedicated to RF spectrum analysis
Bridge	Bridge for long distance, can form mesh
Flex Plus Bridge	FlexConnect + Bridge mode

WLAN Architectures

Autonomous	Autonomous AP, configured manually, all logic on the AP, connects via trunk port,
Lightweight	Split-MAC, AP handles lightweight functions, WLC does everything else, Lightweight AP connects via access port
Cloud-Based	Between Split-MAC and Autonomous, can configure itself, only management/control traffic is sent to the WLC

WLC Deployments

Unified (Hardware in central location)	Max 6000 APs
Cloud-Based (WLC is a VM on a server)	Max 3000 APs
Embedded (WLC in a switch)	Max 200 APs
Mobility Express (WLC in an AP)	Max 100 APs

WLC Ports (Physical)

Service Port	Dedicated management port, must connect to access port (one VLAN)
Distribution System Port	Standard network port, wireless to wired network, can form a LAG
Console Port	Console, RJ45 or USB
Redundancy Port	Connect to another WLC - form HA (High Availability) pair

WLC Interfaces (Logical)

Management	Management traffic (Telnet, SSH, HTTP(S), RADIUS, CAPWAP)
Redundancy Management	To connect and manage the “standby” WLC in the HA pair
AP-Manager	Manage the lightweight APs, source IP for the lightweight APs
Virtual	Communicate with wireless clients - relay DHCP requests, etc.
Service Port	Bound to physical service port, out-of-band management
Dynamic	Map WLAN to VLAN, send wireless to the wired LAN

CAPWAP Tunnels

Control (UDP 5246)	Data (UDP 5247)
Configure APs, control/management operations	Traffic from wireless clients sent through this tunnel
Encrypted by default	Not encrypted by default

Encryption and Integrity Protocols

Protocol	Encryption	Integrity	Used In
TKIP	RC4	MIC	WPA
CCMP	AES Counter Mode	CBC-MAC	WPA2
GCMP	AES Counter Mode	GMAC	WPA3

WPA2/3 Modes

WPA2 Personal	PSK Authentication
WPA2 Enterprise	802.1x + EAP Authentication
WPA3 Personal	SAE to replace PSK handshake (protection against dictionary attacks) + PMF (Protected Management Frames)
WPA3 Enterprise	802.1x + EAP Authentication, Mandatory PMF

Wireless Authentication

Open Authentication	Client sends authentication request, no questions asked
WEP	Auth and encryption (RC4), shared key protocol, AP sends challenge phrase
LEAP	Client provides User and Pass, mutual auth via challenge phrases
EAP-FAST	PAC from server to client, TLS tunnel created, then client auth happens in TLS tunnel
PEAP	Digital certificate to auth the server and establish TLS tunnel, then client auth inside of TLS tunnel
EAP-TLS	Requires a certificate on the AS and every supplicant, TLS tunnel used to exchange encryption key information

WLAN GUI Security

VPN Pass-through	Layer 3	Security for WLANs
Web Authentication	Layer 3	Security for Guest LANs
Web Passthrough	Layer 3	Access for Guest LANs
802.1x	Layer 2	EAP + Dynamic WEP Key
None + EAP Passthrough	Layer 2	Open auth + remote EAP auth
WPA + WPA2	Layer 2	WPA or WPA2
802.1x + CCKM	Layer 2	802.1x clients won't need to reauthenticate when roaming

WLC GUI Tabs

Monitor	WLANs	Controller	Wireless	Security	Management
See AP and Client Summary	Create/edit WLANs, (general, security, QoS, Advanced)	Create/edit interfaces, manage ports	View APs, edit AP modes, radio config., QoS config.	Create/apply CPU ACLs, View AAA settings	View Management Settings, config. management connections

WLAN GUI QoS Settings

Platinum (voice)
Gold (video)
Silver (best effort) (default)
Bronze (background)

WLC CLI Configuration Commands

Allow HTTP connections (disabled by default)	config network webmode {enable | disable}
Allow HTTPS connections (enabled by default)	config network secureweb {enable | disable}
Enable/disable Telnet connections	config network telnet{enable | disable}
Enable/disable SSH connections	config network ssh {enable | disable}
Config. minutes of inactivity before timeout from telnet/SSH sessions (0 = no timeout)	config sessions timeout <timeout>
Config. number of simultaneous Telnet/SSH sessions allowed	config sessions maxsessions <session_num>

Virtualization

VMs and Containers

Virtual Machines	Containers
Require a Hypervisor	Do not require a Hypervisor (shares a kernel)
Has dedicated resources	Share host resources
Requires more overhead	Requires less overhead

Automation and SDN

SD-Access Deployments

Greenfield (Optimal)	Brownfield
Entirely new deployment	Uses existing network
DNA Center configures the underlay	Underlay not configured by DNA Center
All switches and links are Layer 3	Some switches might be Layer 2
IS-IS used as routing protocol	Any routing protocol (recommended IS-IS)
No need for STP	Layer 2 switches still need STP
No need for a FHRP	Layer 3 switches use a FHRP
Access layer switches are default gateway	Distribution switches are default gateway

SD-Access Protocols

Cisco TrustSec	Provides policy control
LISP	Provides the control plane
VXLAN	Provides the data plane

SD-Access Layers

Application Layer	Contains scripts, applications; talks to the Northbound API
Control Layer	Contains the Controller (DNA Center, now called Catalyst Center)
Infrastructure Layer	Contains the network devices that forward messages on the network

SD-Access Architecture

Underlay	The physical network of devices and connections, switches use a separate IP address space than the rest of the enterprise
Overlay	The virtual network built on top of the underlay, uses VXLAN
Fabric	Combination of overlay and underlay, the network as a whole

SD-Access Underlay Switch Roles

Edge Node	Connects to end hosts
Border Node	Connects to devices outside SD-Access domain, ex: WAN routers
Control Node	Uses LISP for control plane functions

CRUD to HTTP Messages

CRUD Acronym	HTTP Message Equivalent	Example
Create	POST	Generate + retrieve token
Read	GET	Retrieve data from object
Update	PUT, PATCH	Add/modify object
Delete	DELETE	Remove an object

HTTP Response Codes

Class of Response	Class Name	Example
1xx	Informational	102 - Processing
2xx	Successful	200 - OK, 201 - Created, 202 - Accepted
3xx	Redirection	301 - Moved Permanently 302 - Moved Temporarily
4xx	Client Error	401 - Unauthorized (auth fail) 404 - Not Found
5xx	Server Error	500 - Internal Server Error

Data Serialization Languages

JSON	XML	YAML
REST API Usage	REST API Usage	REST API Usage (not often)
Whitespace Insignificant	Whitespace Insignificant	Whitespace Significant
Key:value	<key>value</key>	Key:value
“Interface f/01”:{“status”:”up”}	<int>G0/1</int>	- interface: G0/1

REST Authentication

Basic Authentication	Username + Password in HTTP Authorization Header
Bearer Authentication	Token in HTTP Authorization Header - Tokens expire
API Key Authentication	Static Key issued by API - valid until revoked
OAuth 2.0	Grants third parties limited access to resources - access delegation Ex: logging in with Google

Ansible and Terraform

Ansible	Terraform
Agentless	Agentless
Push Model	Push Model
Configuration Management Tool	Infrastructure Provisioning Tool
Procedural	Declarative
Python (Playbooks written in YAML)	Core written in GO, Files written in HCL
Red Hat (Open source)	HashiCorp

Miscellaneous

Hardware

CPU	Management and Control plane operations
ASIC	Data plane operation, processes data traffic for forwarding
TCAM	Stores tables, ACLs, QoS lookups, returns entry to ASIC for forwarding

Formulas

Conversion	Formula	Example
AF to DSCP	8x + 2y (x = Priority, y = Drop Precedence)	AF32 = 8(3) + 2(2) = DSCP 28
CS to DSCP	8(CS value)	CS2 = 8(2) = DSCP 16
Prefix to Usable Hosts (IPv4)	$2^h - 2$ (h = host bits)	/24 = 8 host bits, $2^8 - 2 = 254$ usable hosts
Prefix to Usable Hosts (IPv6)	2^h (h = host bits)	/120 = 8 host bits $2^8 = 256$ usable hosts *IPv6 does not reserve first or last addresses (network and broadcast)
OSPF Interface Cost	Reference Bandwidth / Interface Bandwidth = Cost	Default Reference = 100Mbps 100/1000(GigEthernet) = 1 *Lowest cost can only be 1
EIGRP Metric	Bandwidth (slowest link) + Delay (all links) = Metric What really happens: [(10,000,000 / bandwidth in kbps) + (total delay in usec / 10)] x 256	100 Mbps = 100,000 Kbps Total Delay = 400 usec $10,000,000/100,000 = 100$ $400/10 = 40$ So: $256 \times (100 + 40) = 35,840$ Metric

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